

9 (**long**) A heavy chain of length L and mass per unit length ρ is resting on a table, as sketched in Fig. 8.

(a) The chain is being pulled up by one of its ends with a force F which varies with time t so that the chain moves with constant velocity v . Determine F , as a function of time, and sketch it on a graph. [8]

(b) The chain is lifted up again, but this time by a constant force F .

(i) Derive the differential equation obeyed by the length of chain in the air, z , and use the following trial solution to find $z(t)$:

$$z(t) = a + bt + ct^2 \quad [12]$$

(ii) Hence, find the time it takes for the whole chain to leave the table. [5]

(iii) Determine the minimum value of F needed for the whole chain to leave the table, at least for an instant, and comment on your answer. [5]

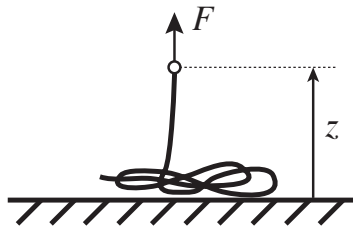


Fig. 8